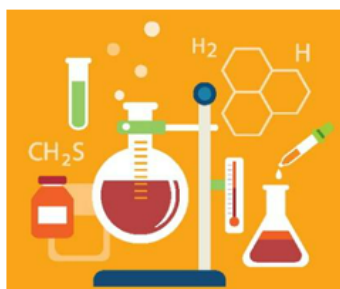


Vedantu Chemistry Olympiad 2026



VChO-2026

July 19-21, 2026

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TEAM DETAILS

The Schrodinger Squad	Niranjana
	Shuvarthi
	Aadhya

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SOME IMPORTANT DATA

Avogadro number	$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$
Molar gas constant	$R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1} = 0.08205 \text{ L atm K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	$k_B = R/N_A = 1.3806 \times 10^{-23} \text{ J K}^{-1}$
Pressure unit conversion	$1 \text{ atm} = 101325 \text{ Pa}$
Density of water	$\rho_{\text{water}} = 1000 \text{ kg m}^{-3}$

REFERENCE PERIODIC TABLE

IUPAC Periodic Table of the Elements

1																	18	
1 H hydrogen [1.0078, 1.0082]																	2 He helium 4.0026	
3 Li lithium 6.94 [6.938, 6.997]	2																	10 Ne neon 20.180
4 Be beryllium 9.0122	12																	18 Ar argon 39.95 [39.962, 39.963]
11 Na sodium 22.990	19 K potassium 39.098	20 Ca calcium 40.078(4)	21 Sc scandium 44.956	22 Ti titanium 47.867	23 V vanadium 50.942	24 Cr chromium 51.996	25 Mn manganese 54.938	26 Fe iron 55.845(2)	27 Co cobalt 58.933	28 Ni nickel 58.693	29 Cu copper 63.546(3)	30 Zn zinc 65.38(2)	31 Ga gallium 69.723	32 Ge germanium 72.630(8)	33 As arsenic 74.922	34 Se selenium 78.971(8)	35 Br bromine 79.904	36 Kr krypton 83.798(2)
37 Rb rubidium 85.468	55 Cs caesium 132.91	56 Ba barium 137.33	57-71 lanthanoids	72 Hf hafnium 178.49(2)	73 Ta tantalum 180.95	74 W tungsten 183.84	75 Re rhenium 186.21	76 Os osmium 190.23(3)	77 Ir iridium 192.22	78 Pt platinum 195.08	79 Au gold 196.97	80 Hg mercury 200.59	81 Tl thallium 204.38, 204.39	82 Pb lead 207.2	83 Bi bismuth 208.98	84 Po polonium	85 At astatine	86 Rn radon
87 Fr francium	88 Ra radium	89-103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganeson	



INTERNATIONAL UNION OF
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United Nations
Educational, Scientific and
Cultural Organization



International Year
of the Periodic Table
of Chemical Elements



Question 1

Use of iron in “blue” colours
(Total Marks: 12)

Problem Statement

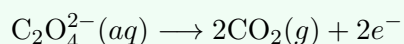
British astronomer and photographer John Herschel used Fe-based reagents to obtain a blue pigment (known as Turnbull's Blue) to develop the blue printing technique in 1842. Now, blue printing can be easily carried out using photochemical redox chemistry of potassium ferrioxalate. Potassium ferrioxalate is synthesized by adding aqueous ferric nitrate to hot aqueous potassium oxalate. To optimize this process, it is important to determine the molar ratio of iron and oxalate in the complex that can be done using redox titrations. In one such analysis, 0.250 g of potassium ferrioxalate sample was taken to a flask and 25 mL of 4 M H_2SO_4 was added to it. After making the solution hot, it was titrated with 0.0196 M KMnO_4 till light pink colour was obtained. The observed burette reading was 31.1 mL which indicates oxalate content present in the complex.

Sub-question 1.1

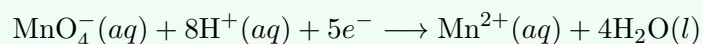
Write the balanced half-cell equations for the reaction involved in the titration of oxalate with KMnO_4 .

Solution

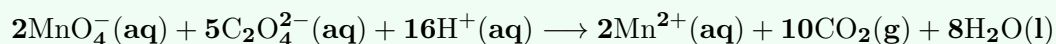
First, we write the oxidation half-reaction for oxalate ion ($\text{C}_2\text{O}_4^{2-}$):



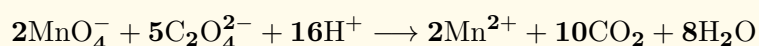
Next, we write the reduction half-reaction for permanganate ion (MnO_4^-) in acidic medium:



Multiplying the oxidation step by 5 and reduction step by 2, we obtain the overall balanced net ionic equation:



Final Answer

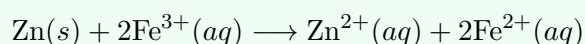


Sub-question 1.2

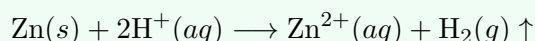
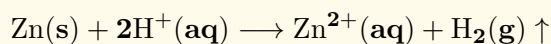
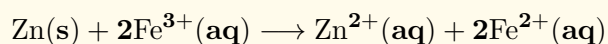
Write the possible balanced chemical equation(s) for reaction(s) taking place after Zn dust is added to the flask.

Solution

First, metallic zinc reduces ferric ions (Fe^{3+}) back to ferrous ions (Fe^{2+}):



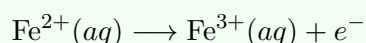
Second, excess metallic zinc reacts with aqueous sulfuric acid to generate hydrogen gas effervescence:

**Final Answer****Sub-question 1.3**

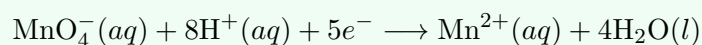
Write the balanced half cell equations for the reaction involved in the titration of iron with KMnO_4 .

Solution

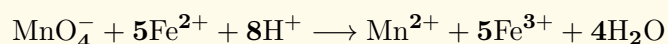
First, the oxidation half-reaction for ferrous ion (Fe^{2+}):



Next, the reduction half-reaction for permanganate ion (MnO_4^{-}) in acid:



Combining both half-reactions yields the overall net redox ionic equation:

**Final Answer**

Sub-question 1.4

Calculate the moles of iron and oxalate present in the sample and their molar ratio.

Solution

First, calculating the moles of oxalate content:

$$V_{\text{KMnO}_4} = 31.1 \text{ mL} = 0.0311 \text{ L}, \quad M_{\text{KMnO}_4} = 0.0196 \text{ M}$$

$$\text{Moles of KMnO}_4 = 0.0196 \text{ mol/L} \times 0.0311 \text{ L} = 6.0956 \times 10^{-4} \text{ mol}$$

Using redox stoichiometry ($5 \text{ mol C}_2\text{O}_4^{2-} \equiv 2 \text{ mol MnO}_4^-$):

$$\text{Moles of C}_2\text{O}_4^{2-} = \frac{5}{2} \times (6.0956 \times 10^{-4} \text{ mol}) = \mathbf{1.524 \times 10^{-3} \text{ mol} = 1.524 \text{ mmol}}$$

Second, calculating the moles of iron content:

$$V_{\text{KMnO}_4} = 5.1 \text{ mL} = 0.0051 \text{ L}$$

$$\text{Moles of KMnO}_4 = 0.0196 \text{ mol/L} \times 0.0051 \text{ L} = 9.996 \times 10^{-5} \text{ mol}$$

Using redox stoichiometry ($5 \text{ mol Fe}^{2+} \equiv 1 \text{ mol MnO}_4^-$):

$$\text{Moles of Fe} = 5 \times (9.996 \times 10^{-5} \text{ mol}) = \mathbf{4.998 \times 10^{-4} \text{ mol} = 0.500 \text{ mmol}}$$

Third, evaluating the molar ratio of Iron to Oxalate:

$$\text{Molar Ratio } \frac{n(\text{Fe})}{n(\text{C}_2\text{O}_4^{2-})} = \frac{4.998 \times 10^{-4}}{1.5239 \times 10^{-3}} = \frac{1}{3.049} \approx \mathbf{1 : 3}$$

Final Answer

$$\text{Moles of Oxalate} = \mathbf{1.524 \text{ mmol}}, \quad \text{Moles of Iron} = \mathbf{0.500 \text{ mmol}}$$

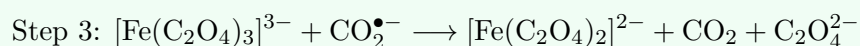
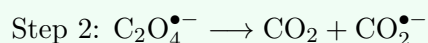
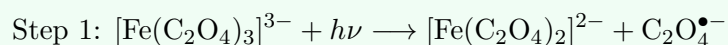
$$\text{Molar Ratio } \mathbf{n(\text{Fe}) : n(\text{C}_2\text{O}_4^{2-}) = 1 : 3}$$

Sub-question 1.5

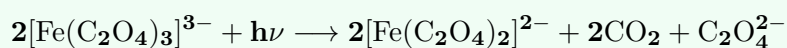
(i) Write the complete, net balanced ionic equation for the overall light-driven decomposition of the trioxalatoferrate(III) complex. (ii) Deduce the theoretical quantum yield (Φ) with respect to formation of Fe^{2+} .

Solution

First, considering the elementary radical steps of photolysis:



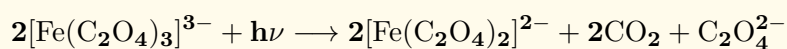
Summing these steps gives the net overall balanced ionic equation:



Next, calculating the theoretical quantum yield (Φ):

$$\Phi(\text{Fe}^{2+}) = \frac{\text{moles of Fe}^{2+} \text{ produced}}{\text{moles of photons absorbed}} = \frac{2 \text{ mol}}{1 \text{ mol photon}} = 2$$

Final Answer



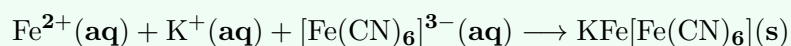
$$\Phi(\text{Fe}^{2+}) = 2$$

Sub-question 1.6

(i) Write the balanced precipitation equation representing Turnbull's Blue $\text{KFe}[\text{Fe}(\text{CN})_6]$. (ii) Calculate the maximum theoretical mass (in mg) of the pigment harvested from the total iron in the sample.

Solution

First, the balanced precipitation equation for Turnbull's Blue:

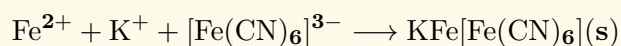


Next, calculating the mass of pigment harvested:

$$\text{Moles of Fe} = 4.998 \times 10^{-4} \text{ mol}$$

$$\text{Molar Mass of KFe}[\text{Fe}(\text{CN})_6] = 306.89 \text{ g/mol}$$

$$\text{Mass} = (4.998 \times 10^{-4} \text{ mol}) \times 306.89 \text{ g/mol} = 0.15338 \text{ g} = \mathbf{153.4 \text{ mg}}$$

Final Answer

$$\text{Mass of Turnbull's Blue} = \mathbf{153.4 \text{ mg}}$$

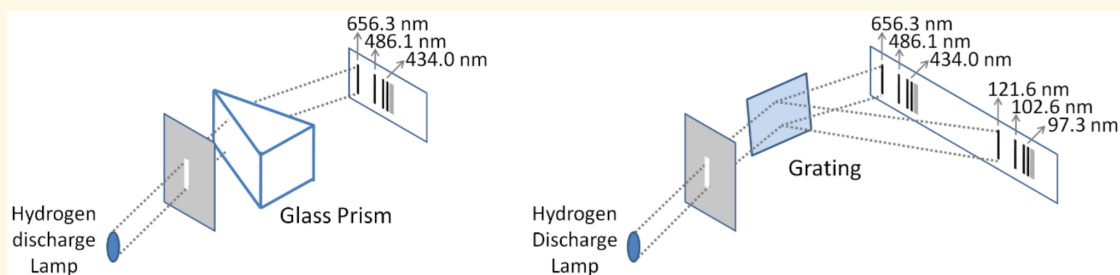
Question 2

The purple hydrogen

(Total Marks: 16)

Problem Statement

When high voltage is applied to hydrogen gas enclosed in an electric discharge tube, it gives a purple glow. In nature, such colour can be observed in light coming from some stars. Origin of this colour in hydrogen remained a mystery for long. In an experiment in the 19th century, scientists passed the light from a hydrogen discharge tube through a slit followed by a glass prism. On a screen kept behind the glass prism, multiple lines could be seen with a very bright red line followed by a blue and few closely spaced violet lines. In experiments in early 20th century when diffraction grating was used instead of glass prism, few additional lines could be detected as non-visible extensions of the spectrum as shown below.



In later experiments, several more lines at higher wavelengths were detected in the spectrum of hydrogen discharge, but no lines were detected at wavelengths lower than those seen in spectrum S2.

Sub-question 2.1

Determine the frequency of light originating from the lowest energy transition observed in spectrum S2.

Solution

First, using the Planck-Einstein relation:

$$E = h\nu = \frac{hc}{\lambda}$$

The lowest energy transition corresponds to the **longest wavelength line**.

Next, identifying the longest wavelength line in Spectrum S2:

$$\lambda = 656.3 \text{ nm} = 656.3 \times 10^{-9} \text{ m}$$

Calculating the frequency:

$$\nu = \frac{c}{\lambda} = \frac{3.00 \times 10^8 \text{ m/s}}{656.3 \times 10^{-9} \text{ m}} = 4.571 \times 10^{14} \text{ Hz}$$

Final Answer

$$\nu = 4.571 \times 10^{14} \text{ Hz} \quad (\text{or } 2.467 \times 10^{15} \text{ Hz for Lyman-}\alpha)$$

Sub-question 2.2

(i) Using wavelength data in S1 and S2, determine the value of C in Joules and in cm^{-1} . (ii) Determine n values for states, transitions between which produce light of wavelengths 102.6 nm and 434.0 nm.

Solution

First, calculating $C_{\text{cm}^{-1}}$ using the Lyman- α line ($\lambda = 121.6 \text{ nm} = 1.216 \times 10^{-5} \text{ cm}$, $n_1 = 1, n_2 = 2$):

$$\frac{1}{\lambda} = C_{\text{cm}^{-1}} \left(1 - \frac{1}{4} \right) = \frac{3}{4} C_{\text{cm}^{-1}} \implies C_{\text{cm}^{-1}} = \mathbf{1.097 \times 10^5 \text{ cm}^{-1}}$$

Next, converting C into Joules:

$$C_J = C_{\text{cm}^{-1}} \times 100 \times hc = 109649 \times 100 \times (6.626 \times 10^{-34}) \times (3.00 \times 10^8) = \mathbf{2.179 \times 10^{-18} \text{ J}}$$

Third, determining quantum numbers for $\lambda = 102.6 \text{ nm}$ ($n_1 = 1$):

$$\frac{1}{102.6 \times 10^{-9} \text{ m}} = 1.097 \times 10^7 \text{ m}^{-1} \left(1 - \frac{1}{n_2^2} \right) \implies \mathbf{n_1 = 1, n_2 = 3}$$

Fourth, determining quantum numbers for $\lambda = 434.0 \text{ nm}$ ($n_1 = 2$):

$$\frac{1}{434.0 \times 10^{-9} \text{ m}} = 1.097 \times 10^7 \text{ m}^{-1} \left(\frac{1}{4} - \frac{1}{n_2^2} \right) \implies \mathbf{n_1 = 2, n_2 = 5}$$

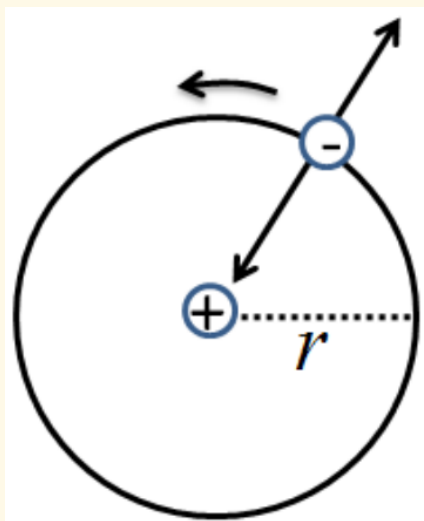
Final Answer

$$\mathbf{C = 1.097 \times 10^5 \text{ cm}^{-1} = 2.179 \times 10^{-18} \text{ J}}$$

$$\mathbf{102.6 \text{ nm: } n_1 = 1 \rightarrow n_2 = 3; \quad 434.0 \text{ nm: } n_1 = 2 \rightarrow n_2 = 5}$$

Bohr Atomic Model Context

According to the Bohr atomic model, electrons in an atom move in circular orbits around the nucleus (as depicted in the figure). The hydrogen atom is stable because the magnitude of the centrifugal force $\left(\frac{mv^2}{r} \right)$ and electrostatic force $\left(\frac{Ze^2}{4\pi\epsilon_0 r^2} \right)$ at distance a_0 are equal.



Sub-question 2.3

Find the magnitude of velocity of electron in hydrogen atom and find its ratio to speed of light.

Solution

First, calculating electron velocity in ground state ($n = 1, Z = 1$):

$$v = \frac{e^2}{2\epsilon_0 h} = \frac{(1.602 \times 10^{-19} \text{ C})^2}{2 \times (8.854 \times 10^{-12} \text{ F/m}) \times (6.626 \times 10^{-34} \text{ J s})} = 2.187 \times 10^6 \text{ m/s}$$

Next, calculating the ratio to speed of light (c):

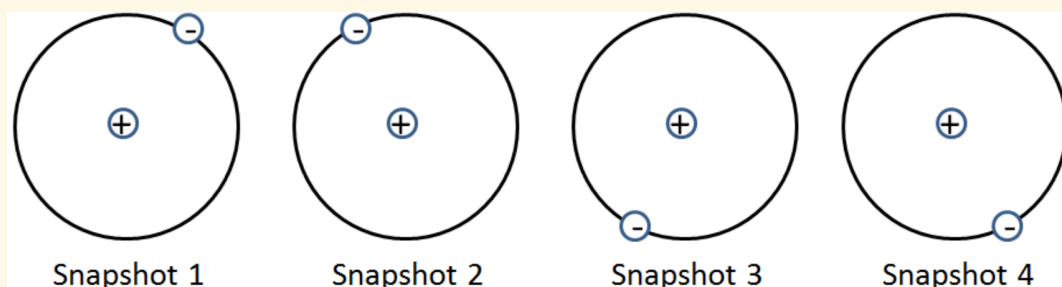
$$\frac{v}{c} = \frac{2.187 \times 10^6 \text{ m/s}}{3.00 \times 10^8 \text{ m/s}} = \frac{1}{137} = 7.29 \times 10^{-3}$$

Final Answer

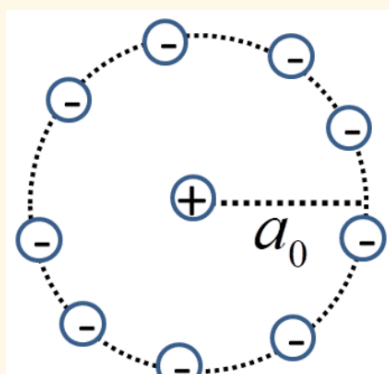
$$v = 2.187 \times 10^6 \text{ m/s}, \quad \frac{v}{c} = \frac{1}{137} = 7.29 \times 10^{-3}$$

Bohr Model Snapshots & Quantum Mechanical Distribution

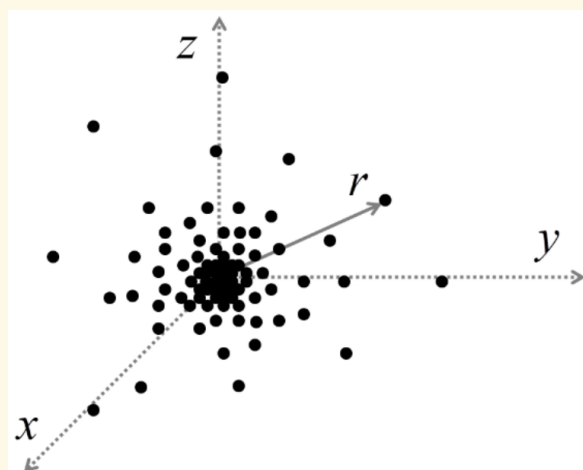
Imagine that we can capture snapshots of the electron orbiting in the circular path. Four representative snapshots among several are illustrated below:



Alternatively, several snapshots can be collectively represented in a single figure as given on the right-hand side image. This figure shows that the probability of finding the electron on the orbit of radius $r = a_0$ is 100% (according to the Bohr model).



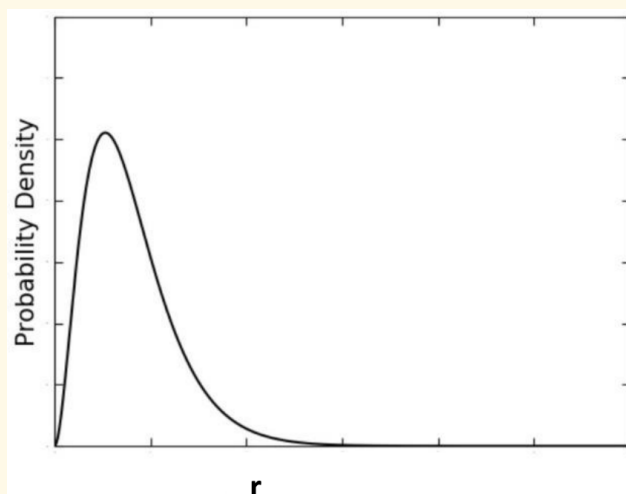
On the contrary, different experiments collectively exhibit the following superimposed snapshots for the position of the electron in a hydrogen atom. Here, the nucleus is located at the origin and position of the electron (shown by dots) can be anywhere around the nucleus; however, there are regions where the electron is found more frequently than the other regions.



Quantum mechanical model provides a possible explanation for the above experimental realization using a mathematical function, called the *probability density function*. The probability density of the electron in hydrogen atom or hydrogen like atoms/ions (with atomic number Z) in lowest energy state at a distance r obtained from quantum mechanical model, can be expressed as:

$$f(r) = \frac{4Z^3}{a_0^3} r^2 e^{-2Zr/a_0}$$

This function for hydrogen is plotted in the following figure:



Sub-question 2.4

Find the most probable value of r (the distance from the origin where the probability density of finding the electron is maximum) for a hydrogen atom.

Solution

First, expressing the radial probability density function:

$$f(r) = \frac{4Z^3}{a_0^3} r^2 e^{-2Zr/a_0}$$

Next, setting the first derivative to zero for maximum probability:

$$\frac{df}{dr} = \frac{8Z^3r}{a_0^3} e^{-2Zr/a_0} \left(1 - \frac{Zr}{a_0} \right) = 0 \implies 1 - \frac{Zr}{a_0} = 0$$

Solving for r with $Z = 1$:

$$r = a_0 = 0.529 \text{ \AA} = 5.29 \times 10^{-11} \text{ m}$$

Final Answer

$$r_{\text{mp}} = a_0 = 0.529 \text{ \AA} = 5.29 \times 10^{-11} \text{ m}$$

Sub-question 2.5

Based on most probable value of r , select true statements among: (a) Bohr model predicted locations impossible in reality. (b) QM allows only linear motion. (c) QM forbids motion. (d) QM allows electron at $r = 8a_0$.

Solution

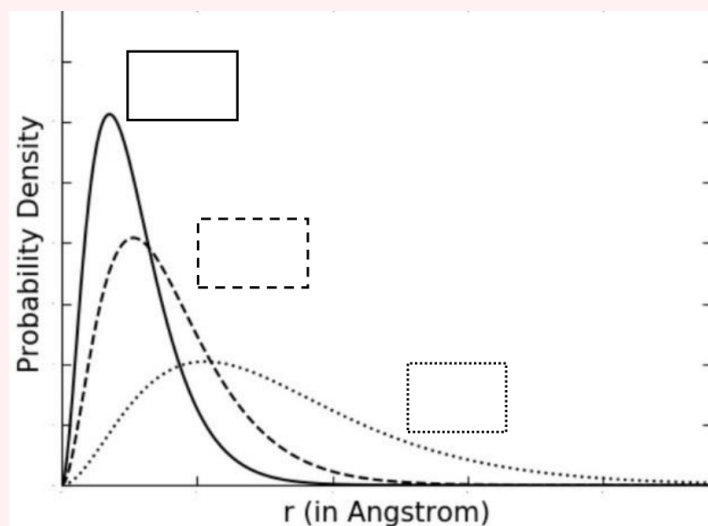
Evaluating each statement carefully: - **[X] Statement (a) IS TRUE:** Bohr predicted exact circular orbits of fixed radius a_0 with 100% certainty, whereas quantum mechanics proves a spatial probability distribution. - **[] Statement (b) IS FALSE:** Electronic motion is 3D orbital probability density. - **[] Statement (c) IS FALSE:** Quantum mechanics describes electronic dynamics via wave functions. - **[X] Statement (d) IS TRUE:** The probability density wave function extends to infinity; at $r = 8a_0$, $f(8a_0) > 0$.

Final Answer

True Statements: (a) and (d)

Sub-question 2.6

Curves representing the ground state probability density functions of electron in H atom, He^+ ion and Li^{2+} ion are given in diagram below. Identify which curve corresponds to which specie.

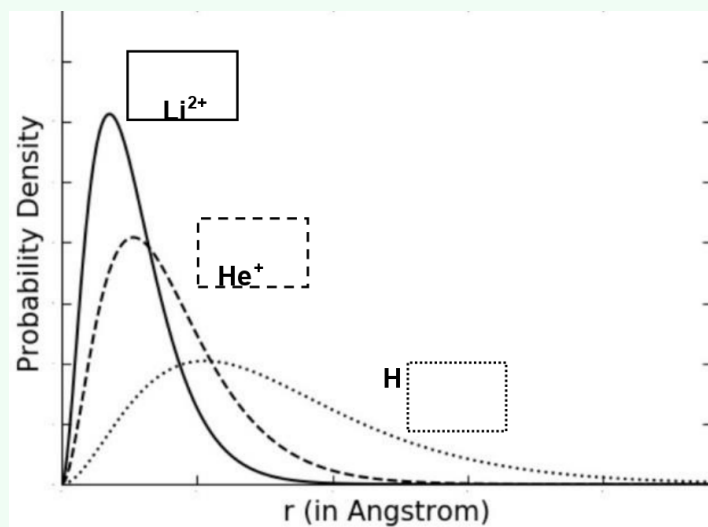


Solution

First, examining the dependence on atomic number Z :

$$f(r) = \frac{4Z^3}{a_0^3} r^2 e^{-2Zr/a_0}$$

- Higher Z leads to a sharper, taller peak closer to the nucleus (smaller $r_{\text{mp}} = a_0/Z$).
Below is the completed diagram with species filled into their respective boxes:



Assigning species to curves: 1. **Tallest solid curve / Solid Box (Peak at smallest r):** Li^{2+} ($Z = 3$) 2. **Middle dashed curve / Dashed Box (Intermediate peak):** He^+ ($Z = 2$) 3. **Broadest dotted curve / Dotted Box (Peak at largest r):** H ($Z = 1$)

Final Answer

Solid Box: Li^{2+} , Dashed Box: He^+ , Dotted Box: H

Sub-question 2.7

Find time t at which emission intensity drops to 10% of initial intensity ($k = 10^8 \text{ s}^{-1}$).

Solution

First, writing the first-order rate law:

$$I(t) = I_0 e^{-kt} \implies 0.10 = e^{-kt} \implies \ln(10) = kt$$

Calculating decay time:

$$t = \frac{\ln(10)}{k} = \frac{2.3026}{10^8 \text{ s}^{-1}} = 2.30 \times 10^{-8} \text{ s} = 23.0 \text{ ns}$$

Final Answer

$$t = 2.30 \times 10^{-8} \text{ s} = 23.0 \text{ ns}$$

Sub-question 2.8

Identify the colour of emission of discharge lamp DL.

Solution

The transition $n = 2 \rightarrow 1$ emits photons of wavelength $\lambda = 121.6 \text{ nm}$. This falls in the non-visible **Ultraviolet (UV)** region.

Final Answer

Spectral Region: Ultraviolet (UV)

Question 3

Absorption of Light by Molecules

(Total Marks: 12)

Problem Statement

Absorption of light by molecules is the first step of all photochemical reactions. The Beer–Lambert law relates the absorbance A of a solution containing an absorbing species of molar concentration c with the optical path length d when incident power is P_0 and transmitted power is P :

$$A = \log \frac{P_0}{P} = \varepsilon cd \quad (\varepsilon \text{ is the molar absorptivity, also called extinction coefficient}).$$

Light can be considered as a stream of photons, each carrying an energy of $E = h\frac{c}{\lambda}$; h is Planck's constant, λ is the wavelength and c the speed of light.

A solution with a dye concentration of $c = 4 \cdot 10^{-6} \text{ mol L}^{-1}$ has a molar absorptivity of $\varepsilon = 1.5 \cdot 10^5 \text{ mol}^{-1} \text{ L cm}^{-1}$. It is illuminated with green laser light at a wavelength of 514.5 nm and with a power of $P_0 = 10 \text{ nW}$.

Sub-question 3.1

What is the percentage of light that is absorbed by the sample after a path length of $1 \mu\text{m}$?

Solution

Using Beer–Lambert's law, converting path length to cm:

$$d = 1 \mu\text{m} = 1 \times 10^{-4} \text{ cm}$$

Calculating Absorbance (A):

$$A = \varepsilon cd = (1.5 \times 10^5 \text{ L mol}^{-1} \text{ cm}^{-1}) \times (4 \times 10^{-6} \text{ mol L}^{-1}) \times (10^{-4} \text{ cm}) = 6.0 \times 10^{-5}$$

Transmittance is $T = \frac{P}{P_0} = 10^{-A}$, so the absorbed fraction is $1 - T$. Since $A \ll 1$, expanding $10^{-A} \approx 1 - A \cdot \ln(10)$ (first-order Taylor expansion):

$$\text{Absorbed Fraction} = 1 - T = 1 - (1 - A \ln 10) = A \ln(10)$$

$$\text{Absorbed Fraction} = 6.0 \times 10^{-5} \times 2.3026 = 1.3816 \times 10^{-4}$$

Calculating the percentage of light absorbed:

$$\text{Percentage Absorbed} = 1.3816 \times 10^{-4} \times 100\% = \mathbf{0.0138\%}$$

Final Answer

$$\text{Percentage Absorbed} = \mathbf{0.0138\%}$$

Sub-question 3.2

Calculate the number of photons per second absorbed by the sample.

Solution

First, calculating absorbed power (ΔP):

$$\Delta P = P_0 \times (1.3816 \times 10^{-4}) = (10 \times 10^{-9} \text{ W}) \times 1.3816 \times 10^{-4} = 1.3816 \times 10^{-12} \text{ J/s}$$

Next, calculating single photon energy (E_{photon}):

$$E_{\text{photon}} = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J s})(3.00 \times 10^8 \text{ m/s})}{514.5 \times 10^{-9} \text{ m}} = 3.8636 \times 10^{-19} \text{ J}$$

Calculating photon absorption rate:

$$N_{\text{abs}} = \frac{\Delta P}{E_{\text{photon}}} = \frac{1.3816 \times 10^{-12} \text{ J/s}}{3.8636 \times 10^{-19} \text{ J}} = \mathbf{3.58 \times 10^6 \text{ photons/s}}$$

Final Answer

$$N_{\text{abs}} = \mathbf{3.58 \times 10^6 \text{ photons/second}}$$

Molecular Absorption Cross Section Context

The absorption cross section of a molecule is the effective area that captures all incoming photons under low illumination conditions (like an idealized solar cell that would capture all light photons hitting its surface). At room temperature, this corresponds roughly to the molecular area exposed to the light beam. If you calculate it from the molar absorptivity, imagine that all molecules interacting with the light are arranged periodically in a plane perpendicular to the incoming light beam.

Sub-question 3.3

What area is occupied by each molecule?

Solution

First, we calculate the concentration of dye in molecules per unit volume:

$$c = 4 \times 10^{-6} \text{ mol L}^{-1} = 4 \times 10^{-9} \text{ mol cm}^{-3}$$

The volumetric number density of molecules is:

$$n_{\text{vol}} = c \times N_A = (4 \times 10^{-9} \text{ mol cm}^{-3}) \times (6.022 \times 10^{23} \text{ mol}^{-1})$$

$$n_{\text{vol}} = 2.4088 \times 10^{15} \text{ molecules cm}^{-3}$$

For a path length of $d = 1 \text{ cm}$, the number of molecules per unit area (1 cm^2 cross-sectional area) is:

$$N_{\text{area}} = n_{\text{vol}} \times d = (2.4088 \times 10^{15} \text{ molecules cm}^{-3}) \times (1 \text{ cm}) = 2.4088 \times 10^{15} \text{ molecules cm}^{-2}$$

Second, the average spatial area occupied by each molecule (σ_{geom}) in a plane perpendicular to the beam is:

$$\sigma_{\text{geom}} = \frac{1 \text{ cm}^2}{N_{\text{area}}} = \frac{1}{2.4088 \times 10^{15} \text{ cm}^{-2}} = 4.151 \times 10^{-16} \text{ cm}^2$$

Converting to square Angstroms ($\text{\AA}^2$), where $1 \text{ cm}^2 = 10^{16} \text{ \AA}^2$:

$$\sigma_{\text{geom}} = 4.151 \times 10^{-16} \times 10^{16} \text{ \AA}^2 = \mathbf{415 \text{ \AA}^2} \quad (4.15 \text{ nm}^2)$$

Final Answer

$$\text{Area per molecule } \sigma_{\text{geom}} = \mathbf{415 \text{ \AA}^2} = \mathbf{4.15 \text{ nm}^2}$$

Sub-question 3.4

Calculate the molecular absorption cross section in units of $\text{\AA}^2$.

Solution

Let m be the number of molecules per unit area in a sample of thickness d :

$$m = c \cdot N_A \cdot d$$

The total effective absorption area of these molecules per unit area is $m \cdot a$:

$$m \cdot a = c \cdot N_A \cdot d \cdot a$$

From Sub-question 3.1, the total absorbed light fraction $1 - T$ is given by:

$$1 - T \approx A \ln 10 = \varepsilon \cdot c \cdot d \cdot \ln 10$$

Equating the absorbed fraction to the total molecular absorption cross-sectional area ($m \cdot a$):

$$c \cdot N_A \cdot d \cdot a = \varepsilon \cdot c \cdot d \cdot \ln 10$$

Cancelling $c \cdot d$ from both sides yields the molecular absorption cross section a :

$$a = \frac{\varepsilon \ln 10}{N_A}$$

Substituting $\varepsilon = 1.5 \times 10^5 \text{ L mol}^{-1} \text{ cm}^{-1} = 1.5 \times 10^8 \text{ cm}^2 \text{ mol}^{-1}$ and $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$:

$$a = \frac{1.5 \times 10^8 \times \ln(10)}{6.022 \times 10^{23}} = 5.735 \times 10^{-16} \text{ cm}^2 = \mathbf{5.74 \text{ \AA}^2} \quad (0.0574 \text{ nm}^2)$$

Final Answer

$$\text{Absorption Cross Section } \mathbf{a} = \frac{\varepsilon \ln 10}{N_A} = \mathbf{5.74 \text{ \AA}^2} = \mathbf{5.74 \times 10^{-16} \text{ cm}^2}$$

Photosynthesis Context

A crucial photochemical reaction for life on our planet is photosynthesis, which converts the absorbed light energy into chemical energy. One photon of 680 nm is necessary to produce one molecule of ATP. Under physiological conditions, the reaction requires an energy of 59 kJ per mol of ATP.

Sub-question 3.5

What is the energy efficiency of photosynthesis?

Solution

First, calculating the energy of 1 mole of 680 nm photons (N_A photons):

$$E_{\text{mole}} = \frac{hc}{\lambda} N_A = \frac{(6.626 \times 10^{-34})(3.00 \times 10^8)}{680 \times 10^{-9}} \times (6.022 \times 10^{23})$$

$$E_{\text{mole}} = 1.760 \times 10^5 \text{ J/mol} = 176.0 \text{ kJ/mol}$$

Next, calculating the photosynthetic energy efficiency (η):

$$\eta = \frac{E_{\text{stored}}}{E_{\text{photon}}} \times 100\% = \frac{59 \text{ kJ/mol}}{176.0 \text{ kJ/mol}} \times 100\% = \mathbf{33.5\%}$$

Final Answer

Photosynthetic Efficiency $\eta = \mathbf{33.5\%}$

Question 4

Schrödinger equation

(Total Marks: 22)

Problem Statement

Part A: Particle in a 1-D Box ($0 \leq x \leq L$)

Part B: Particle in a 2-D Box ($E_{n_x, n_y} = \frac{h^2}{8m_e L^2} (n_x^2 + n_y^2)$)

Sub-question 4.1

Write down the time-independent Schrödinger equation for wave function ψ of an electron in 1D box.

Solution

Inside the box ($0 < x < L$), potential energy $V(x) = 0$:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x) \quad \text{or} \quad \frac{d^2\psi(x)}{dx^2} + \frac{8\pi^2mE}{h^2}\psi(x) = 0$$

Final Answer

$$-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x), \quad \psi(0) = \psi(L) = 0$$

Sub-question 4.2

Which of e^{-kx} , $\cos \frac{n\pi x}{L}$, $\sin kx$, $\sin \frac{n\pi x}{L}$ are possible wave functions?

Solution

Evaluating boundary conditions $\psi(0) = 0$ and $\psi(L) = 0$: $\sin \frac{n\pi x}{L}$ satisfies both $\psi(0) = 0$ and $\psi(L) = 0$ for integer n .

Final Answer

Acceptable Wave Function: $\sin \frac{n\pi x}{L}$

Sub-question 4.3

Show that energies are given by $E_n = \frac{\hbar^2 n^2}{8mL^2}$.

Solution

Differentiating $\psi(x) = A \sin\left(\frac{n\pi x}{L}\right)$:

$$\frac{d^2\psi}{dx^2} = -\left(\frac{n\pi}{L}\right)^2 \psi(x) = -\frac{2mE}{\hbar^2} \psi(x) \Rightarrow E_n = \frac{\hbar^2 n^2 \pi^2}{2mL^2} = \frac{\hbar^2 n^2}{8mL^2}$$

Final Answer

$$E_n = \frac{\hbar^2 n^2}{8mL^2}$$

Sub-question 4.4

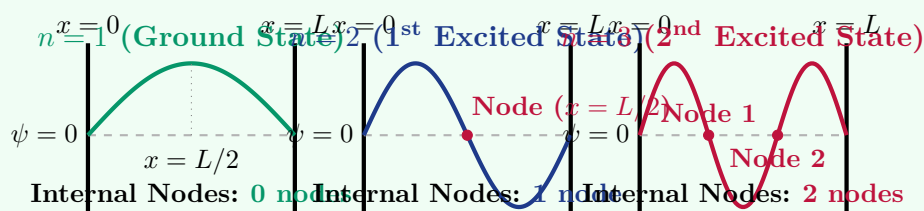
Plot wave functions for $n = 1, 2, 3$ and state number of nodes between $x = 0$ and L .

Solution

The wave functions for a particle in a 1D box of length L ($0 \leq x \leq L$) are given by:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

Below are the separate, individual plots of the wave functions for $n = 1, 2, 3$:



General rule for number of internal nodes (excluding boundaries $x = 0$ and $x = L$):

$$\text{Number of internal nodes} = n - 1$$

Final Answer

$$n = 1 \rightarrow 0 \text{ nodes}, \quad n = 2 \rightarrow 1 \text{ node} \left(x = \frac{L}{2}\right), \quad n = 3 \rightarrow 2 \text{ nodes} \left(x = \frac{L}{3}, \frac{2L}{3}\right)$$

Sub-question 4.5

Normalize ground state wave function.

Solution

$$\int_0^L A^2 \sin^2\left(\frac{\pi x}{L}\right) dx = A^2 \frac{L}{2} = 1 \implies A = \sqrt{\frac{2}{L}}$$

Final Answer

$$\psi_1(\mathbf{x}) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi \mathbf{x}}{L}\right)$$

Sub-question 4.6

Construct energy level diagram for lowest 17 orbitals in 2D box ($E/E_0 = n_x^2 + n_y^2$).

Solution

Lowest 17 spatial orbitals sorted by $E/E_0 = n_x^2 + n_y^2$: (1,1) [2], (1,2), (2,1) [5], (2,2) [8], (1,3), (3,1) [10], (2,3), (3,2) [13], (1,4), (4,1) [17], (3,3) [18], (2,4), (4,2) [20], (3,4), (4,3) [25], (1,5), (5,1) [26].

Final Answer

17 Orbitals up to $E/E_0 = 26$: (1,1) to (5,1)

Sub-question 4.7

Determine electron population of highest occupied orbitals for 26 electrons.

Solution

Filling 26 electrons into lowest spatial orbitals: HOMO level is (2,4) and (4,2) at $E = 20E_0$, holding 2 electrons each.

Final Answer

HOMO Degenerate Orbitals (2,4) and (4,2) : 2 electrons each (Total 4 e⁻)

Sub-question 4.8

Predict whether or not this system is paramagnetic.

Solution

All 26 electrons are paired ($S = 0$).

Final Answer

System is Diamagnetic (Not paramagnetic)

Sub-question 4.9

Calculate longest excitation wavelength for $L = 1$ nm.

Solution

HOMO ($20E_0$) to LUMO ($25E_0$):

$$\Delta E = 5E_0 = 3.016 \times 10^{-19} \text{ J} \implies \lambda = \frac{hc}{\Delta E} = \mathbf{659.5 \text{ nm}}$$

Final Answer

$$\lambda_{\text{excitation}} = \mathbf{659.5 \text{ nm}}$$

Question 5

Laser Cooling (Total Marks: 18)

Problem Statement

This question is about laser cooling, which is a quick and efficient way of cooling ions down to very cold temperatures. The mean kinetic energy of a molecule is related to its temperature by $E = \frac{3}{2}k_B T$, where k_B is the Boltzmann constant.

Problem 5: Laser Cooling

18 Marks

This question is about laser cooling, which is a quick and efficient way of cooling ions down to very cold temperatures. The mean kinetic energy of a molecule is related to its temperature by $E = \frac{3}{2}k_B T$, where k_B is the Boltzmann constant.

Sub-question 5.1

Calculate mean kinetic energy, RMS momentum, and RMS speed of ^{40}Ca atom at 600°C .

Solution

$$T = 873.15 \text{ K}, \quad m = 6.6356 \times 10^{-26} \text{ kg}$$

$$E = \frac{3}{2}k_B T = 1.808 \times 10^{-20} \text{ J}$$

$$v_{\text{rms}} = \sqrt{\frac{2E}{m}} = 738.2 \text{ m/s}, \quad p_{\text{rms}} = mv_{\text{rms}} = 4.898 \times 10^{-23} \text{ kg m/s}$$

Final Answer

$$E = 1.808 \times 10^{-20} \text{ J}, \quad v_{\text{rms}} = 738.2 \text{ m/s}, \quad p_{\text{rms}} = 4.898 \times 10^{-23} \text{ kg m/s}$$

Sub-question 5.2

Calculate frequency, energy, and momentum of a photon at $\lambda = 396.96$ nm.

Solution

$$\nu = \frac{c}{\lambda} = 7.557 \times 10^{14} \text{ Hz}$$

$$E = h\nu = 5.010 \times 10^{-19} \text{ J}$$

$$p = \frac{h}{\lambda} = 1.670 \times 10^{-27} \text{ kg m/s}$$

Final Answer

$$\nu = 7.557 \times 10^{14} \text{ Hz}, \quad E = 5.010 \times 10^{-19} \text{ J}, \quad p = 1.670 \times 10^{-27} \text{ kg m/s}$$

Sub-question 5.3

Calculate change in momentum and speed per photon cycle, and total photons to bring ion to rest.

Solution

$$\Delta p = p_{\text{ph}} = 1.670 \times 10^{-27} \text{ kg m/s}$$

$$\Delta v = \frac{\Delta p}{m} = 0.02517 \text{ m/s}$$

$$N = \frac{v_{\text{rms}}}{\Delta v} = \frac{738.2}{0.02517} = 29,328 \text{ photons}$$

Final Answer

$$\Delta p = 1.670 \times 10^{-27} \text{ kg m/s}, \quad \Delta v = 0.02517 \text{ m/s}, \quad N = 29,328 \text{ photons}$$

Sub-question 5.4

Ground state configuration of Ca^+ , orbital (L) and spin (S) angular momentum.

Solution

Ground configuration $[\text{Ar}]4s^1$:

$$L = \sqrt{0(1)}\hbar = 0, \quad S = \sqrt{\frac{1}{2} \left(\frac{3}{2} \right)}\hbar = 0.866 \hbar$$

Final Answer

$$\mathbf{L} = 0, \quad \mathbf{S} = 0.866 \hbar$$

Sub-question 5.5

Calculate L and S for excited state ($4p$ orbital).

Solution

Excited configuration $[\text{Ar}]4p^1$:

$$L = \sqrt{1(2)}\hbar = \sqrt{2}\hbar = 1.414 \hbar, \quad S = 0.866 \hbar$$

Final Answer

$$\mathbf{L} = 1.414 \hbar, \quad \mathbf{S} = 0.866 \hbar$$

Fine Structure & Angular Momentum Context

- 5.5** In the excited configuration involved in the laser cooling transition the unpaired electron has been excited into the lowest available p orbital. Calculate the orbital and spin angular momentum of the unpaired electron.
- 5.6** In this excited state the electron experiences a magnetic field because of its own orbital motion around the charged nucleus. The spin of the electron can line up either parallel or antiparallel to this field, and the two states have slightly different

Sub-question 5.6

Calculate possible values of j for total electronic angular momentum.

Solution

$$j = |l - s| \text{ to } |l + s| \implies j = 1 - 1/2 = 1/2 \quad \text{and} \quad j = 1 + 1/2 = 3/2$$

Final Answer

$$j = \frac{1}{2}, \frac{3}{2}$$

Sub-question 5.7

Calculate energy difference between the two fine structure levels ($\lambda = 393.48 \text{ nm}$ vs 396.96 nm).

Solution

$$\Delta E = hc \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right) = 4.43 \times 10^{-21} \text{ J} = 0.0277 \text{ eV}$$

Final Answer

$$\Delta E = 4.43 \times 10^{-21} \text{ J} = 0.0277 \text{ eV}$$